

Quiz 07

Instructions: Answer the questions with precise definitions. Your definitions should either be stated exactly as in the lecture or textbook, or be clearly equivalent to them. Having the “right idea” but not a precise formulation will not receive credit.

Problem 1. What is the probability density of a normal random variable Y with mean μ and variance σ^2 ?

Problem 2. Let Z be a standard normal random variable. For what number x is

$$\mathbb{P}[-x \leq Z \leq x] \approx .95?$$